encountered these problems only because we did not understand the true rules of the game; the new rules tell us that there is a limit to how finely we can probe the universe—and, in a real sense, a limit to how finely our conventional notion of distance can even be applied to the ultramicroscopic structure of the cosmos. The supposed pernicious spatial fluctuations are now seen to have arisen in our theories because we were unaware of these limits and were thus led by a point-particle approach to grossly overstep the bounds of physical reality.

Given the apparent simplicity of this solution for overcoming the problem between general relativity and quantum mechanics, you might wonder why it took so long for someone to suggest that the point-particle description is merely an idealization and that in the real world elementary particles do have some spatial extent. This takes us to our second point. Long ago, some of the greatest minds in theoretical physics, such as Pauli, Heisenberg, Dirac, and Feynman, did suggest that nature's constituents might not actually be points but rather small undulating "blobs" or "nuggets." They and others found, however, that it is very hard to construct a theory, whose fundamental constituent is not a point particle, that is nonetheless consistent with the most basic of physical principles such as conservation of quantum-mechanical probability (so that physical objects do not suddenly vanish from the universe, without a trace) and the impossibility of faster-than-light-speed transmission of information. From a variety of perspectives, their research showed time and again that one or both of these principles were violated when the point-particle paradigm was discarded. For a long time, therefore, it seemed impossible to find a sensible quantum theory based on anything but point particles. The truly impressive feature of string theory is that more than twenty years of exacting research has shown that although certain features are unfamiliar, string theory does respect all of the requisite properties inherent in any sensible physical theory. And furthermore, through its graviton pattern of vibration, string theory is a quantum theory containing gravity.

The More Precise Answer

The rough answer captures the essence of why string theory prevails where previous point-particle theories failed. And so, if you like, you can go on to the next section without losing the logical flow